Pragmatics-Aware Token Selection and Scheduling for Generative Semantic Communication over Time-Varying Channels

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***Abstract*—Generative semantic communication (GSC) achieves efficient transmission by converting multi-modal content into prompt. Yet, existing methods treat the prompt as a monolithic unit and do not differentiate individual token importance under time-varying channels. We propose a token-level pragmatics-aware selection and adaptive scheduling method that constructs a unified pragmatic metric from self-attention structural weights, semantic directional consistency, and generative causal gradient sensitivity, with a semantic coupling matrix capturing inter-token dependencies. Token selection is solved via Lagrangian relaxation with online dual-variable updates and joint semantic–channel bandwidth allocation for multi-terminal scenarios. Experiments on ScienceQA demonstrate consistently leading VQA accuracy (Acc) and CLIPScore (CS) across four channel models. It achieves the best Acc and CS in 13 out of 16 settings, and its advantage scales with bandwidth pressure, and achieves accuracies of 0.81–0.91 and reaches a Pareto-improving point that no baseline reaches. Its advantage is pronounced under low SNR and adverse channels.**

***Index Terms*—Dynamic prompt pruning, Generative semantic communication, Semantic coupling, Time-varying channel scheduling, Token-level pragmatic metric.**

I. Introduction

THE explosive growth of image and video traffic from intelligent terminals has pushed conventional bit-accurate communication toward its information-theoretic ceiling. Semantic communication (SC) [1] represents a paradigm shift that transmits task-relevant semantics rather than raw bits, treating the encoder and decoder as cooperative inference modules rather than waveform reproducers. Early SC realizations adopt deep joint source-channel coding (DeepJSCC) to map signals into feature vectors [2], with notable progress in single-modal tasks and task-oriented multi-user scenarios [3]. The feature dimensionality required for high-dimensional multi-modal content remains substantial. Generative semantic communication (GSC) reduces this cost by one to two orders of magnitude [4], [5]: the transmitter uses a vision-language model (VLM) to convert multi-modal inputs into compact natural-language prompts, and the receiver reconstructs semantic content via a diffusion model or a large language model (LLM). Within a prompt, however, tokens contribute unevenly to the output, and their relative importance further varies with the instantaneous channel state.

Existing studies have approached this problem from complementary perspectives. Information-bottleneck criteria [6] offer principled tools for semantic feature selection, while fading-robust DeepJSCC [7] and online-adaptive transform-based transmission [8] improve resilience under channel fluctuations. Multi-terminal SC [9], joint user association with bandwidth allocation [10], and large AI model-empowered multimodal SC [11] broaden the applicability of SC systems. Despite this progress, three gaps remain. On the one hand, semantic importance is typically modeled at the feature or prompt level, leaving intra-prompt token heterogeneity unexploited. On the other hand, channel-adaptive transmission and semantic selection are usually optimized in isolation, so that the selection threshold cannot follow instantaneous capacity. Additionally, multi-terminal bandwidth allocation rarely reflects per-prompt semantic urgency, so that terminals with urgent content can be starved while those with redundant content waste channel resources.

To bridge these gaps, this letter presents a new pragmatics-aware token-level selection and adaptive scheduling method for time-varying multi-terminal GSC. The main contributions are as follows:

- We design a multi-dimensional pragmatic contribution metric that fuses self-attention structural weights, semantic directional consistency, and generative causal gradient sensitivity, together with a semantic coupling matrix that captures bidirectional inter-token dependencies.
- We design a semantic benefit density driven token selection strategy, in which Lagrangian relaxation yields an efficient decision rule and a two-stage approximation combining greedy selection with coupling-aware local optimization to reduce the complexity from exponential to polynomial.
- We design a joint scheduling scheme for time-varying multi-terminal channels, in which per-terminal online dual updates adapt the selection threshold to instantaneous capacity and a centralized bandwidth allocation couples terminals through semantic priority and channel state.

Extensive experiments on ScienceQA across four channel models demonstrate that the proposed method achieves leading VQA accuracy and CLIPScore in 13 out of 16 settings,

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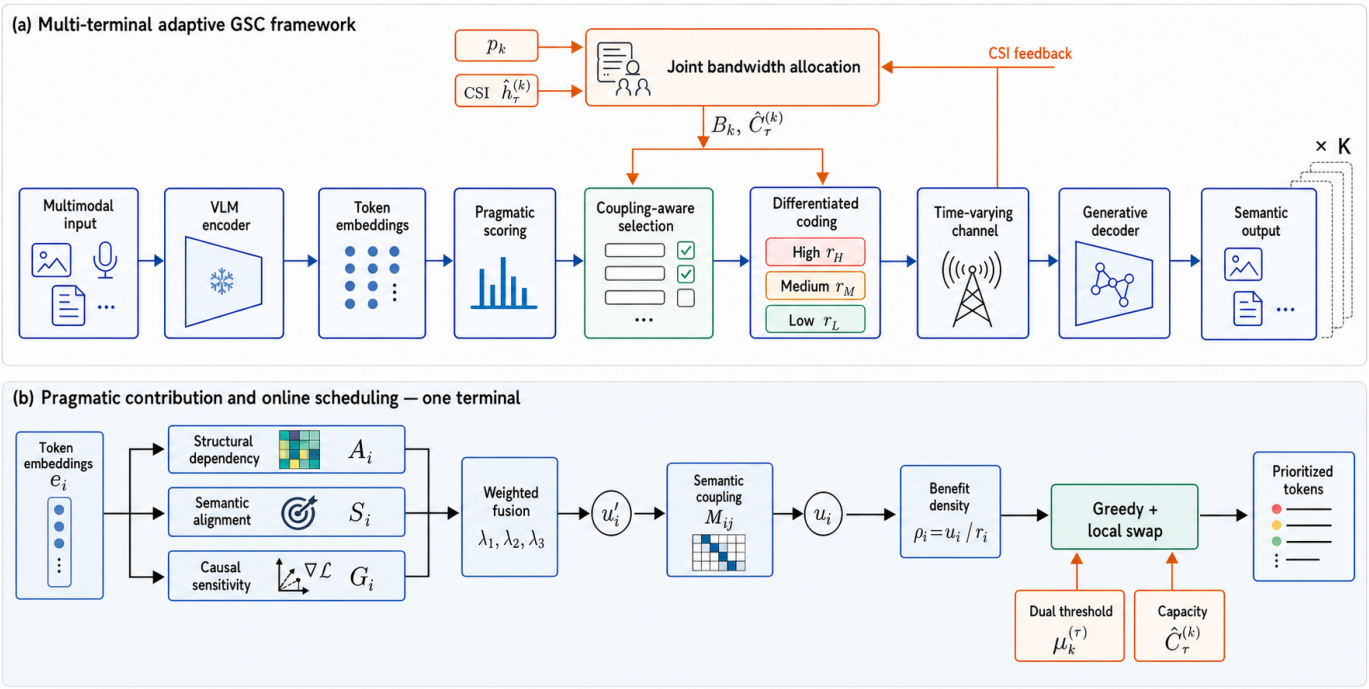


Fig. 1. Multi-terminal pragmatics-aware GSC framework. Each of K terminals runs VLM encoding, pragmatic contribution evaluation, and dynamic pruning in parallel; a Joint Bandwidth Allocation module aggregates per-terminal priorities $\{p_k\}$ and CSI $\{\hat{h}_\tau^{(k)}\}$ to assign capacities $\{\hat{C}_\tau^{(k)}\}$. The lower part depicts per-terminal pragmatic evaluation, where A_i, S_i, G_i are fused into u_i' and refined by the coupling matrix M_{ij} to yield u_i.

sustains accuracy above 0.78 during deep fades, 0.23 higher than the prompt-level baseline, and attains a Pareto-improving fairness–accuracy trade-off in multi-terminal scenarios.

II. System Model and Problem Formulation

A. Prompt Token Representation and Channel Transmission Model

As illustrated in Fig. 1, we consider a system, where the transmitter converts multi-modal inputs into a prompt sequence via a VLM and transmits the encoded tokens over a wireless channel; the receiver then reconstructs semantic content using an LLM. Let $\mathcal{P} = \{w_1, w_2, \ldots, w_N\}$ denote the prompt sequence generated by the transmitter, with each token $\boldsymbol{e}_i \in \mathbb{R}^d$ embedded. A binary selection variable $x_i \in \{0, 1\}$ determines the transmitted subset $\mathcal{T} = \{w_i \mid x_i = 1\}$.

Each selected token is transmitted after differentiated channel coding. Based on pragmatic contribution, tokens are classified into high-, medium-, and low-priority levels and encoded at rates r_H, r_M and r_L ($r_H > r_M > r_L$), respectively. High-priority tokens receive strong error protection, whereas low-priority tokens conserve bandwidth. The total transmission cost is $R = \sum_{i=1}^{N} x_i r_i$, where $r_i \in \{r_H, r_M, r_L\}$ is determined by the priority level.

After channel coding, the token subset is transmitted through a time-varying wireless channel. Let $h_\tau \in \mathbb{C}$ denote the channel coefficient at time slot τ. All channels incorporate channel state information (CSI) estimation error $\hat{h}_\tau = h_\tau + e_\tau$ ($e_\tau \sim \mathcal{CN}(0, \sigma_e^2)$) to model imperfect CSI available at the transmitter via a low-rate feedback link, as well as random packet loss. Token-level packet loss abstracts residual errors after channel decoding: when the decoded symbol group corresponding to a token fails its integrity check, the entire token is marked as lost. The instantaneous channel capacity is $C_\tau = B \log_2(1 + |\hat{h}_\tau|^2 \bar{\gamma})$, where $\bar{\gamma}$ is the average SNR and C_τ is normalized to channel symbols per slot, so that the selected tokens must satisfy $\sum_{i=1}^{N} x_i r_i \le C_\tau$.

B. Pragmatic Utility Maximization and Multi-Terminal Extension

To preserve as much semantics as possible under limited channel capacity, a token subset with the highest transmission value must be selected from the prompt. We introduce a pragmatic utility function u_i to quantify the contribution of token w_i to the semantic fidelity of the generated output. The token selection problem seeks to find the selection variables $\{x_i\}$ that maximize the total pragmatic utility $\sum_{i=1}^{N} x_i u_i$, subject to the rate constraint $\sum_{i=1}^{N} x_i r_i \le C_\tau$. This is a capacity-constrained 0–1 knapsack problem. The additive objective is a first-order approximation of the true utility function that accounts for semantic coupling. Non-additive inter-token interactions are compensated through the subsequent semantic coupling refinement, keeping the overall complexity at $O(N^2)$.

In multi-terminal scenarios, K terminals share a total bandwidth B, with the allocation $\{B_k\}$ satisfying $\sum_{k=1}^{K} B_k = B$. The instantaneous capacity of terminal k is $C_\tau^{(k)} = B_k \log_2(1 + |\hat{h}_\tau^{(k)}|^2 \bar{\gamma})$, and each terminal independently solves its token selection problem under $C_\tau^{(k)}$. Service fairness is measured by the Jain index $J = (\sum_k a_k)^2 / (K \sum_k a_k^2)$, where a_k denotes the long-term average bandwidth allocated to terminal k.

III. Token-Level Pragmatics-Aware Selection Method

A. Multi-Dimensional Pragmatic Contribution Metric

The pragmatic contribution of a token to the generated output is influenced by multiple factors. We characterize it from three complementary aspects: structural dependency, semantic representation, and generation process.

1) Structural dependency: Self-attention captures dynamic inter-token dependencies. Suppose that the model has H attention heads, and $\alpha_{j,i}^{(h)}$ denotes the normalized attention weight from token j to token i in head h. The self-attention structural weight is defined as

$$A_i = \frac{1}{H}\sum_{h=1}^{H}\left(\frac{1}{N}\sum_{j=1}^{N}\alpha_{j,i}^{(h)}\right), \tag{1}$$

where A_i measures how frequently token i is attended to by other tokens, reflecting its role as an information hub.

2) Semantic representation: A token's pragmatic value also depends on how closely its meaning aligns with the overall task semantics. Let $\boldsymbol{g} = \sum_{i=1}^{N} A_i \boldsymbol{e}_i / \sum_{i=1}^{N} A_i$ denote a task-semantic direction vector aggregated from all token embeddings, where A_i serves as the aggregation weight so that weakly attended tokens are prevented from dominating $\boldsymbol{g}$. The semantic directional consistency of token i is then defined as

$$S_i = \frac{\boldsymbol{e}_i^{\top}\boldsymbol{g}}{\|\boldsymbol{e}_i\|_2\|\boldsymbol{g}\|_2}, \tag{2}$$

where S_i measures the alignment between token i and the global task semantics.

3) Generation process: The generative causal gradient sensitivity is defined as

$$G_i = \left\|\frac{\partial \mathcal{L}}{\partial \boldsymbol{e}_i}\right\|_2, \tag{3}$$

where $\mathcal{L}$ is the autoregressive negative log-likelihood loss of the transmitter's VLM. The gradient computation requires only one additional backward pass without receiver feedback. G_i serves as a proxy for receiver-side task performance sensitivity: tokens with large G_i exert stronger influence on the described content and are more likely to affect receiver performance.

B. Unified Pragmatic Contribution Function and Semantic Coupling Model

The three metrics differ in scale and range, and cannot be compared directly. After normalizing each component, we define the unified pragmatic contribution function as

$$u_i' = \lambda_1 \tilde{A}_i + \lambda_2 \tilde{S}_i + \lambda_3 \tilde{G}_i, \tag{4}$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and the weights λ_1, λ_2 and λ_3 are determined by grid search on a validation set.

To capture inter-token contextual dependencies (e.g., the attribute binding between "red" and "cube"), we introduce a semantic coupling matrix based on multi-head self-attention:

$$M_{ij} = \frac{1}{H}\sum_{h=1}^{H}\frac{\alpha_{i,j}^{(h)} + \alpha_{j,i}^{(h)}}{2}. \tag{5}$$

The context-aware pragmatic contribution is then defined as

$$u_i = u_i' + \beta \sum_{j\neq i} M_{ij} u_j', \tag{6}$$

where β is a coupling regulation parameter satisfying $\beta < 1/\max_i \sum_{j\neq i} M_{ij}$ to ensure the coupling does not overturn the dominance of the base contribution u_i'. Note that u_i is precomputed on the full token set; when some tokens are not selected for transmission, an overestimation bias $\delta_i = \beta \sum_{j\notin\mathcal{T}} M_{ij} u_j'$ arises, which is bounded by $\delta_i \leq \beta|\bar{\mathcal{T}}| \cdot \bar{M}$ since $u_j' \leq 1$ after normalization, with $\bar{M}$ the average coupling strength over unselected tokens.

C. Lagrangian Relaxation and Semantic Benefit Density Decision

Directly solving this 0–1 knapsack problem incurs exponential complexity. To enable efficient optimization, Lagrangian relaxation transforms this constrained problem into an unconstrained form:

$$L(\{x_i\},\mu) = \sum_{i=1}^{N} x_i(u_i - \mu r_i) + \mu C_\tau, \tag{7}$$

where $\mu \geq 0$ is the Lagrangian multiplier. Because L is separable in $\{x_i\}$, the decision criterion reduces to

$$x_i^* = \begin{cases} 1, & \text{if } \rho_i > \mu; \\ 0, & \text{otherwise,} \end{cases} \tag{8}$$

where $\rho_i = u_i/r_i$ is the semantic benefit density. Since r_i varies with the priority level, ρ_i jointly reflects pragmatic value and coding overhead. Greedy selection by sorting ρ_i in descending order and accumulating the coding rate, combined with local optimization, yields an overall complexity of $O(N^2)$. In multi-terminal scenarios, each terminal k maintains its own dual variable μ_k, updated online via $\mu_k^{(\tau+1)} = [\mu_k^{(\tau)} + \eta(\sum_i x_i^{(k,\tau)} r_i^{(k)} - \hat{C}_\tau^{(k)})]^+$ with a constant step size η and $x_i^{(k,\tau)}$ given by (8), which keeps μ_k within an $O(\eta)$ neighborhood of the optimum to maintain tracking capability under channel variations.

D. Online Semantic Scheduling under Time-Varying Channels

The per-token selection rule in (8) acts independently on each token, but does not account for the effect of the actual selected set on the coupling refinement. In time-varying wireless environments, we design a two-stage approximation to refine the selection result, as summarized in Algorithm 1.

The first stage selects tokens satisfying $\rho_i^{(k)} > \mu_k^{(\tau)}$ in descending order of $\rho_i^{(k)}$, subject to the capacity budget, to form $\mathcal{T}_0^{(k)}(\tau)$, following the decision rule in Section III-C. In the second stage, marginal gains are re-evaluated based on the current selected set. For each unselected token j, the insertion gain is $\Delta_j^+ = u_j' + 2\beta \sum_{m\in\mathcal{T}_0^{(k)}} M_{jm} u_m'$; for the token l in the set with the lowest benefit density, the removal loss Δ_l^- is computed analogously. A swap is performed when $\Delta_j^+ > \Delta_l^-$ and the rate constraint remains feasible, for at most $T_{\max}$ rounds.

Algorithm 1 Online Semantics-Aware Token Scheduling per Terminal k

Require: $\{u_i^{(k)}\}$, $\{u_i'^{(k)}\}$, $\{r_i^{(k)}\}$, $\{M_{ij}^{(k)}\}$, allocated B_k from coordinator, step size η, max swap rounds $T_{\max}$, $\mu_k^{(0)} = 0$

for each time slot τ **do**
  $\hat{C}_\tau^{(k)} \leftarrow B_k \log_2(1 + |\hat{h}_\tau^{(k)}|^2 \bar{\gamma})$
  $\rho_i^{(k)} \leftarrow u_i^{(k)} / r_i^{(k)}$, $\forall i$; sort descending
  $\mathcal{T}_0^{(k)}(\tau) \leftarrow \emptyset$, $R^{(k)} \leftarrow 0$
  for each token i in sorted order **do**
    if $\rho_i^{(k)} > \mu_k^{(\tau)}$ **and** $R^{(k)} + r_i^{(k)} \le \hat{C}_\tau^{(k)}$ **then**
      $\mathcal{T}_0^{(k)} \leftarrow \mathcal{T}_0^{(k)} \cup \{i\}$, $R^{(k)} \leftarrow R^{(k)} + r_i^{(k)}$
    end if
  end for
  for $t = 1$ to $T_{\max}$ **do**
    $j^* \leftarrow \arg\max_{j \notin \mathcal{T}_0^{(k)}} \Delta_j^+$; $\quad l^* \leftarrow \arg\min_{i \in \mathcal{T}_0^{(k)}} \rho_i^{(k)}$
    if $\Delta_{j^*}^+ > \Delta_{l^*}^-$ **and** $R^{(k)} - r_{l^*}^{(k)} + r_{j^*}^{(k)} \le \hat{C}_\tau^{(k)}$ **then**
      $\mathcal{T}_0^{(k)} \leftarrow \mathcal{T}_0^{(k)} \cup \{j^*\} \setminus \{l^*\}$
    else
      break
    end if
  end for
  $\mu_k^{(\tau+1)} \leftarrow [\mu_k^{(\tau)} + \eta(\sum_{i:\rho_i^{(k)} > \mu_k^{(\tau)}} r_i^{(k)} - \hat{C}_\tau^{(k)})]^+$
  output $\mathcal{T}_0^{(k)}(\tau)$ for transmission in slot τ
end for

The above scheduling runs per terminal under its allocated B_k. To assign $\{B_k\}$, a centralized coordinator collects each terminal's priority $p_k = \frac{1}{N_k} \sum_i u_i^{(k)} / r_i^{(k)}$ and CSI estimate $|\hat{h}_\tau^{(k)}|$, and then sets

$$B_k = B \cdot \frac{p_k \cdot |\hat{h}_\tau^{(k)}|}{\sum_{j=1}^K p_j \cdot |\hat{h}_\tau^{(j)}|}, \tag{9}$$

which steers bandwidth toward terminals with both high semantic value and favorable channels. Coordination overhead is only $2K$ scalars per slot.

IV. Experimental Evaluation

A. Experimental Setup

Experiments are conducted on ScienceQA [12], which contains 21,208 questions (about half with images) spanning natural science, social science, and language science. Each sample comprises a question, candidate options, an optional image, textual context, and expert explanations, making it a suitable benchmark for evaluating end-to-end GSC performance on knowledge-intensive tasks. We stratified-sample 5,000 image-containing multi-modal instances and employ five-fold cross-validation. The transmitter uses BLIP-2 to generate prompts and perform token-level pruning; the receiver infers with Flan-T5-Large; semantic similarity is evaluated using CLIP ViT-L/14. The prompt length is truncated to $N \le 64$, and differentiated coding rates are $r_H/r_M/r_L = 48/32/16$ (channel symbols per token), with priorities assigned by the u_i terciles. Pragmatic weights are $(\lambda_1, \lambda_2, \lambda_3) = (0.25, 0.30, 0.45)$ (validation-set grid search), $\beta = 0.25$, $T_{\max} = 8$, and $\eta = 0.1$. To assess robustness, we evaluate the method under four channel models: Rayleigh fading, Rician fading, frequency-selective fading, and DeepMIMO I1 [13] ray-tracing. Channel parameters: SNR $\in \{0, 15\}$ dB, channel estimation error $\sigma_e^2 = 0.01$, packet loss rate $p = 0.05$, and Rayleigh temporal correlation $\alpha = 0.9$.

TABLE I
VQA Accuracy and CLIPScore Comparison under Four Channel Models

Channel	SNR	Prompt-SC		Adapt.-SC		IB-Sel.		LLMLin.		Proposed	
		Acc	CS	Acc	CS	Acc	CS	Acc	CS	Acc	CS
Rayleigh	0	0.71	0.65	0.74	0.69	0.78	0.71	0.79	0.73	**0.82**	**0.77**
	15	0.89	0.86	0.91	0.88	0.92	0.87	**0.94**	**0.91**	0.93	0.90
Rician	0	0.74	0.68	0.76	0.71	0.79	0.73	0.81	0.75	**0.84**	**0.78**
	15	0.90	0.87	0.92	0.89	0.92	**0.90**	**0.94**	0.89	**0.94**	0.89
Freq.-Sel.	0	0.66	0.59	0.72	0.66	0.73	0.68	0.75	0.68	**0.79**	**0.74**
	15	0.83	0.80	0.87	0.83	0.86	0.81	0.89	0.84	**0.91**	**0.86**
DeepMIMO	0	0.65	0.58	0.71	0.64	0.73	0.67	0.72	0.65	**0.78**	**0.71**
	15	0.81	0.76	0.85	0.81	0.84	0.79	0.86	0.81	**0.89**	**0.84**

The baselines cover three categories: whole-prompt transmission (Prompt-SC [5]), which transmits the entire prompt without token differentiation; channel-adaptive feature transmission (Adaptive-SC [8]); and token-level pruning, including IB-Select, which adapts the information-bottleneck criterion of [6] from feature-level encoding to token-level selection, and LLMLingua [14] using perplexity-based compression, with the last two being the most directly comparable. Evaluation metrics are VQA accuracy (Acc) and CLIPScore (CS).

B. Results and Analysis

Table I presents end-to-end performance under four channel models. The proposed method achieves the best Acc and CS in 13 out of 16 settings. The advantage is the most pronounced under low SNR and adverse channels: at SNR = 0 dB, it outperforms the strongest baseline by 0.03–0.05 in Acc and 0.03–0.06 in CS, with the largest Acc gain (0.05) under DeepMIMO ray-tracing and CS gain (0.06) under frequency-selective fading, validating that token-level differentiated protection and coupling-aware selection are the most advantageous under constrained channel resources. Under frequency-selective fading, where channel diversity stresses naive pruning, the advantage persists across both SNR levels (0.02–0.06). At SNR = 15 dB in Rayleigh and Rician channels, all token-level methods approach the performance ceiling; the proposed method ties with LLMLingua while substantially outperforming prompt-level and feature-level baselines.

Fig. 2 examines robustness under time-varying Rayleigh fading at SNR = 15 dB. Although LLMLingua matches the proposed method at peak slots ($t = 9$), its accuracy collapses during deep fades (slots 5, 12, 19), while the proposed method sustains Acc at 0.78–0.79, 0.23 higher than Prompt-SC's 0.55 at the same fading troughs. The proposed method's standard deviation (0.041) and max-min spread (0.13) are both 40% lower than LLMLingua's, confirming that the online dual-variable update and the bandwidth allocation guard mechanism collaboratively flatten the accuracy curve under channel volatility. Robustness extends to harsher situations: under $\sigma_e^2 = 0.05$ and $p = 0.1$, the proposed method's Acc drop

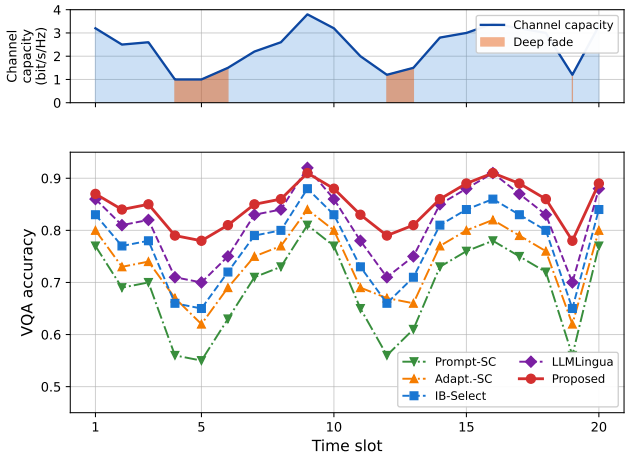


Fig. 2. VQA accuracy over 20 time slots under a time-varying Rayleigh channel at SNR = 15 dB. The upper subplot shows instantaneous channel capacity with deep fading regions (slots 5, 12, and 19) shaded. The proposed method maintains the smallest accuracy fluctuation and sustains above 0.78 during deep fades.

TABLE II
ABLATION STUDY OF THE PROPOSED METHOD UNDER RAYLEIGH CHANNEL

Configuration	SNR = 0 dB		SNR = 15 dB	
	Acc	CS	Acc	CS
w/o $\tilde{A}$ (attention)	0.79	0.73	0.92	0.89
w/o $\tilde{S}$ (semantic direction)	0.76	0.69	0.91	0.87
w/o $\tilde{G}$ (gradient sensitivity)	0.72	0.65	0.90	0.86
w/o coupling matrix $\mathbf{M}$	0.74	0.68	0.90	0.87
w/o differentiated coding	0.73	0.66	0.91	0.88
w/o local swap (Stage-2)	0.77	0.71	0.92	0.89
w/o online dual μ_k	0.78	0.72	0.92	0.89
Full	**0.82**	**0.77**	**0.93**	**0.90**

is 0.04, less than half of Prompt-SC's 0.09, demonstrating the inherent fault tolerance of token-level selection.

Table II reports ablation under Rayleigh fading. At SNR = 0 dB, every component contributes a non-trivial 0.03–0.10 Acc drop when ablated: removing $\tilde{G}_i$ causes the largest drop (0.10), identifying the autoregressive gradient norm as the most critical pragmatic dimension; differentiated coding and the coupling matrix $\mathbf{M}$ rank next (0.09 and 0.08). At SNR = 15 dB, ablation effects narrow because the channel can transmit most tokens unprotected; the gap re-opens precisely when budget tightens, which is the intended operating regime of the proposed framework.

TABLE III
MULTI-TERMINAL BANDWIDTH ALLOCATION COMPARISON

Strategy	$K = 4$		$K = 10$		$K = 20$	
	Acc	Jain	Acc	Jain	Acc	Jain
Equal	0.86	**1.000**	0.79	**1.000**	0.72	**1.000**
Max-CSI	0.88	0.71	0.81	0.58	0.74	0.46
Proportional	0.83	0.95	0.75	0.92	0.68	0.88
Semantic-only	0.89	0.83	0.83	0.78	0.77	0.71
Joint (Proposed)	**0.91**	0.89	**0.86**	0.84	**0.81**	0.79

Table III evaluates multi-terminal resource allocation under Rayleigh fading at SNR = 15 dB. The proposed Joint strategy achieves the highest Acc at all K, and its advantage scales with bandwidth pressure: from 0.02 over Semantic-only at $K = 4$ to 0.04 at $K = 20$, while outperforming Max-CSI by 0.07 and Equal by 0.09 at $K = 20$. This confirms that pairing semantic priority p_k with channel gain $|\hat{h}_\tau^{(k)}|$ is essential when neither signal alone suffices. Notably, the Joint strategy preserves a Jain fairness of 0.79–0.89, dominating Max-CSI (0.46–0.71) on fairness while approaching Semantic-only on accuracy, yielding a Pareto-improving point that no baseline can reach. Regarding complexity, the per-slot latency is 37.9 ms, well below the 100 ms vehicular constraint; N scaling from 16 to 128 raises latency from 8 to 84 ms, and sparse coupling matrices reduce complexity to $O(Ns)$ ($s \ll N$ couplings per token) for long-context scenarios.

V. CONCLUSION

This letter has advanced GSC from prompt-level to token-level pragmatic granularity. A unified pragmatic metric fuses self-attention structure weights, semantic direction consistency, and generative causal gradient sensitivity, with a coupling-aware refinement capturing inter-token dependencies. This metric drives a Lagrangian two-stage selection coupled to joint bandwidth allocation across K terminals. The method leads in 13 of 16 channel settings, sustains accuracy above 0.78 during deep fades, and reaches a Pareto-improving fairness–accuracy trade-off. Future work will replace grid-searched weights with adaptive learning, extend the dual update to cross-slot reinforcement learning, and investigate sparse coupling for long-context regimes.

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